

SmartRoute AI: An Explainable Risk-Aware Deep Reinforcement Learning Framework for Dynamic Last-Mile Delivery Assignment

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Project guide
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Introduction

- Last-mile delivery is the final, costliest and most difficult stage of the logistics chain.
- Key decision: which available delivery partner should serve each arriving order?
- Poor choices increase distance, delay deliveries and overload some partners while others stay idle.
- The Hungarian Algorithm is optimal for a fixed snapshot, but orders and traffic keep changing.
- SmartRoute AI is a DRL framework that adds explicit risk awareness and plain-language explanations.
- Trained and evaluated in a software simulator, so no physical hardware is required.

Abstract

Last-mile delivery is a critical stage of logistics where assigning delivery orders to suitable delivery partners efficiently is challenging due to changing traffic conditions, delivery deadlines, partner availability, distance, and operational risks. This project proposes SmartRoute AI, an explainable and risk-aware deep reinforcement learning framework for dynamic last-mile delivery assignment. The system continuously analyzes delivery requests, delivery-partner availability, traffic conditions, distance, estimated travel time, and delivery priorities. A Deep Reinforcement Learning (DRL) agent learns to select suitable delivery partners for incoming orders based on the current logistics environment. The framework incorporates risk factors such as traffic congestion, delayed deliveries, excessive travel distance, and partner workload into the decision-making process. Unlike conventional assignment methods, the proposed system dynamically adapts to changes in real-time delivery conditions. The Hungarian Algorithm can be integrated as a baseline or optimization component for comparing assignment performance. An explainable AI layer provides understandable reasons for each assignment decision, such as lower travel time, reduced risk, or better workload balance. The system aims to minimize delivery delays, travel distance, and operational risk while improving partner utilization. A simulation-based environment can be used to train and evaluate the DRL model without requiring physical hardware. Performance can be measured using delivery time, assignment accuracy, total distance, workload distribution, and risk reduction. The proposed framework provides a scalable software-based solution for intelligent, adaptive, and explainable last-mile delivery management.

Base Paper

Reference	Vásconez et al. (2024), Smart Cities 7, 1109–1125, “Smart Delivery Assignment through Machine Learning and the Hungarian Algorithm”
Contribution	ML-estimated cost matrices combined with the Hungarian algorithm for optimal order–courier assignment
Methodology	Haversine distance, linear and quadratic polynomial regression, then Hungarian optimisation
Dataset	MAX Delivery (Barcelona), 7,707 completed orders, Jan–Jun 2023
Results	Polynomial regression gave the lowest total time (130,747.82 min vs 147,891.23 min); linear + Hungarian gave 34,957.74 km total distance
Limitation	Static cost matrix; no risk, workload balance or explanations; RL named only as future work

Literature Survey

Work	Methodology	Limitation	Our contribution
Boysen et al. [1]; Dantzig & Ramser [15]	Last-mile delivery and vehicle routing optimisation	Mostly offline, known demand	Dynamic, online assignment
Kuhn [2] (Hungarian)	Optimal one-to-one matching on a cost matrix	One snapshot; no learning of future effects	Used as baseline; DRL learns a policy
Nazari [5]; Kool [6]	Neural / RL vehicle routing	Route construction; no risk or explanation	Continuous assignment with risk
Xu [7]; Lin [8]	Learned ride-hailing dispatch and fleet management	Efficiency focus; little per-decision explanation	Explanation layer for each decision
García [13]; Achiam [14]; Puiutta [11]; Heuillet [12]	Safe and explainable RL	Rarely applied to delivery assignment	Risk-aware, explainable dispatch

System Architecture

Input layer

- Delivery requests (location, deadline, priority)
- Partner status (position, load, availability)
- Traffic and distance data (travel-time estimates)

Decision layer

- State builder
- Risk module (congestion, delay, distance, workload)
- DRL agent
- Hungarian baseline

Output layer

- Assignment decision
- Explainable AI layer (reasons)
- Performance metrics

Modelled as a Markov decision process: state = orders, partners, traffic; action = assign an order to a partner; reward = on-time bonus minus weighted delay, distance, risk and workload imbalance.

Module Flow

1

Simulator

Generates orders, partners and time-varying traffic (Gym interface)

2

State + Risk

Builds state; scores congestion, delay, distance, workload

3

DRL Agent

DQN or PPO scores each order–partner pair and picks the best

4

Baselines

Hungarian Algorithm and nearest-partner rule on the same snapshot

5

Explain

Reason per assignment: travel time, risk, workload balance

Planned stack: Python, PyTorch, Stable-Baselines3; SHAP / LIME for finer attribution.

Evaluation Plan

Criterion	Measure	Purpose
Delivery time	Mean time per order; on-time rate	Speed and reliability
Assignment accuracy	Agreement with Hungarian solution	Quality vs optimal baseline
Total distance	Distance travelled by all partners	Cost efficiency
Workload distribution	Spread of orders across partners	Utilisation and fairness
Risk reduction	Mean risk score of assignments	Operational safety
Explanation	Reason for every assignment	Transparency

Status: design stage. Quantitative results will be reported after the simulation experiments (light, moderate, heavy loads; normal and congested traffic).

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